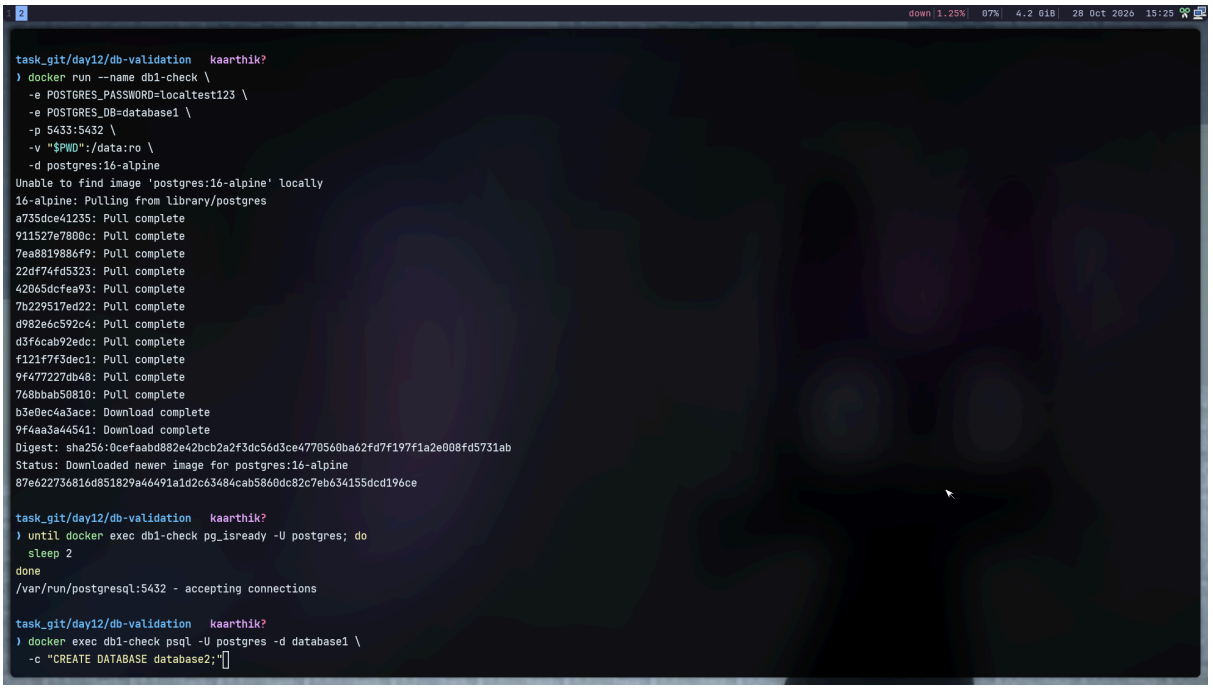
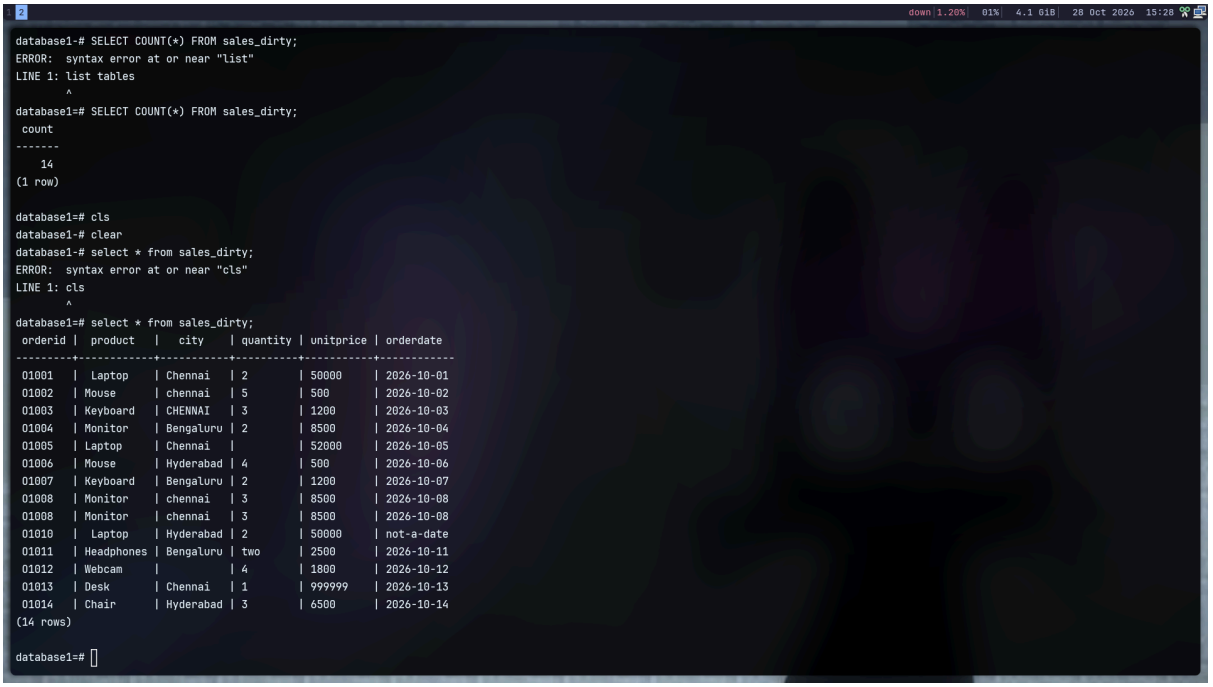
Pulling postgres docker images



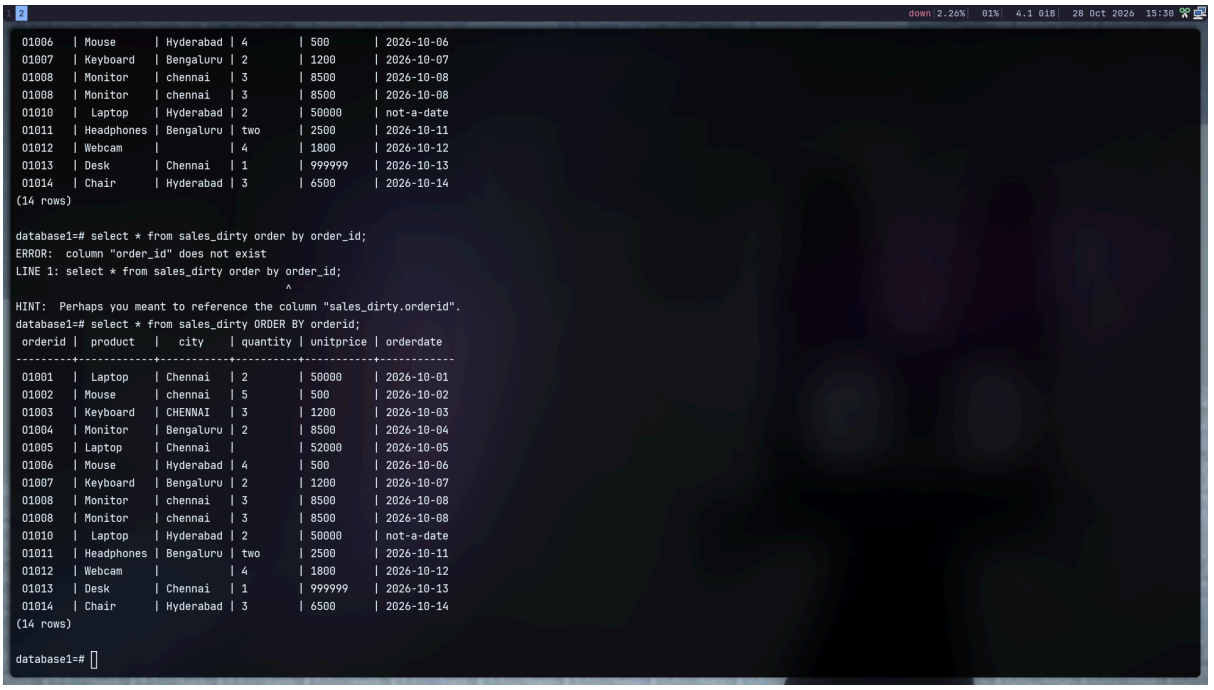
Schema for database 1

```
CREATE TABLE sales_dirty (
    orderid TEXT,
    product TEXT,
    city TEXT,
    quantity TEXT,
    unitprice TEXT,
    orderdate TEXT);
```

Data have been imported from csv file to db with this cmd (\copy sales_dirty FROM '/data/sales_dirty.csv' WITH (FORMAT csv, HEADER true, NULL 'NaN')



1. **Retrieve all the records by ascending order_id**



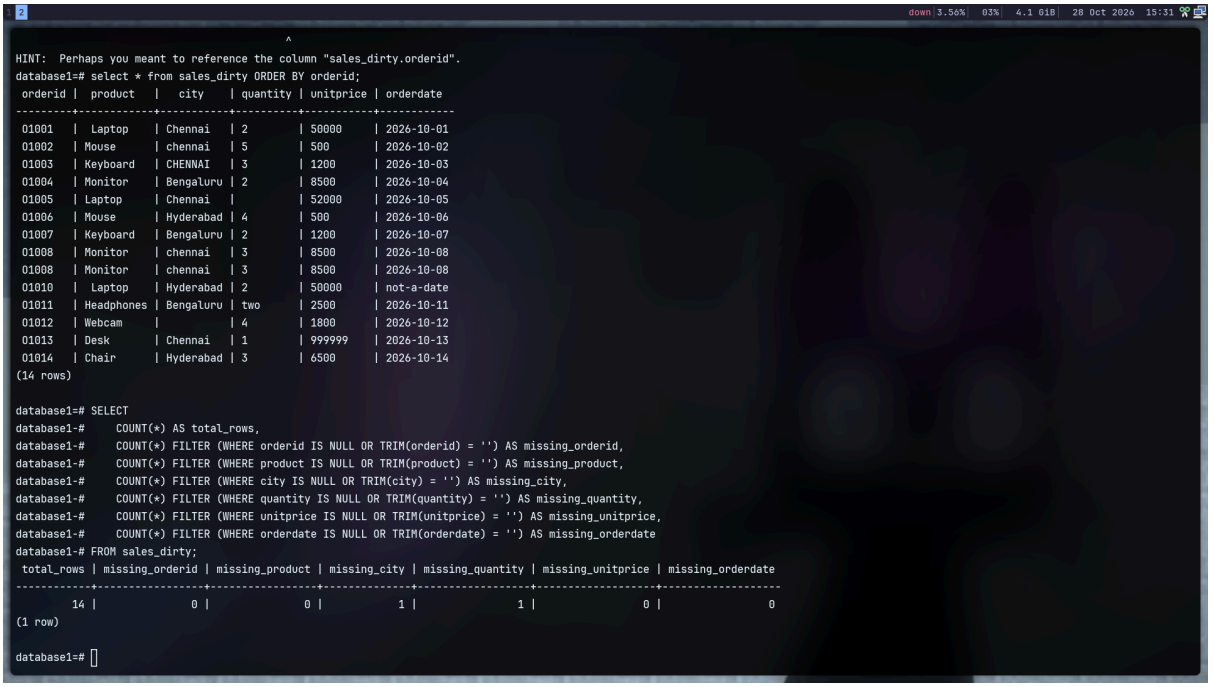
```
 01006   | Mouse      | Hyderabad | 4        | 500       | 2026-10-06
 01007   | Keyboard   | Bengaluru | 2        | 1200      | 2026-10-07
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01010   |  Laptop    | Hyderabad | 2        | 50000     | not-a-date
 01011   | Headphones | Bengaluru | two      | 2500      | 2026-10-11
 01012   | Webcam     |           | 4        | 1800      | 2026-10-12
 01013   | Desk       | Chennai   | 1        | 999999    | 2026-10-13
 01014   | Chair      | Hyderabad | 3        | 6500      | 2026-10-14
(14 rows)

database1=# select * from sales_dirty order by order_id;
ERROR:  column "order_id" does not exist
LINE 1: select * from sales_dirty order by order_id;
                                           ^
HINT:  Perhaps you meant to reference the column "sales_dirty.orderid".
database1=# select * from sales_dirty ORDER BY orderid;
 orderid |  product   |   city    | quantity | unitprice | orderdate
---------+------------+-----------+----------+-----------+-----------
 01001   |  Laptop    | Chennai   | 2        | 50000     | 2026-10-01
 01002   | Mouse      | chennai   | 5        | 500       | 2026-10-02
 01003   | Keyboard   | CHENNAI   | 3        | 1200      | 2026-10-03
 01004   | Monitor    | Bengaluru | 2        | 8500      | 2026-10-04
 01005   | Laptop     | Chennai   |          | 52000     | 2026-10-05
 01006   | Mouse      | Hyderabad | 4        | 500       | 2026-10-06
 01007   | Keyboard   | Bengaluru | 2        | 1200      | 2026-10-07
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01010   |  Laptop    | Hyderabad | 2        | 50000     | not-a-date
 01011   | Headphones | Bengaluru | two      | 2500      | 2026-10-11
 01012   | Webcam     |           | 4        | 1800      | 2026-10-12
 01013   | Desk       | Chennai   | 1        | 999999    | 2026-10-13
 01014   | Chair      | Hyderabad | 3        | 6500      | 2026-10-14
(14 rows)

database1=#
```

2. **Check Completeness -> Validate: Missing values should be reported separately for every column.**

Query : SELECT
    COUNT(*) AS total_rows,
    COUNT(*) FILTER (WHERE orderid IS NULL OR TRIM(orderid) = '') AS missing_orderid,
    COUNT(*) FILTER (WHERE product IS NULL OR TRIM(product) = '') AS missing_product,
    COUNT(*) FILTER (WHERE city IS NULL OR TRIM(city) = '') AS missing_city,
    COUNT(*) FILTER (WHERE quantity IS NULL OR TRIM(quantity) = '') AS missing_quantity,
    COUNT(*) FILTER (WHERE unitprice IS NULL OR TRIM(unitprice) = '') AS missing_unitprice,
    COUNT(*) FILTER (WHERE orderdate IS NULL OR TRIM(orderdate) = '') AS missing_orderdate
FROM sales_dirty;



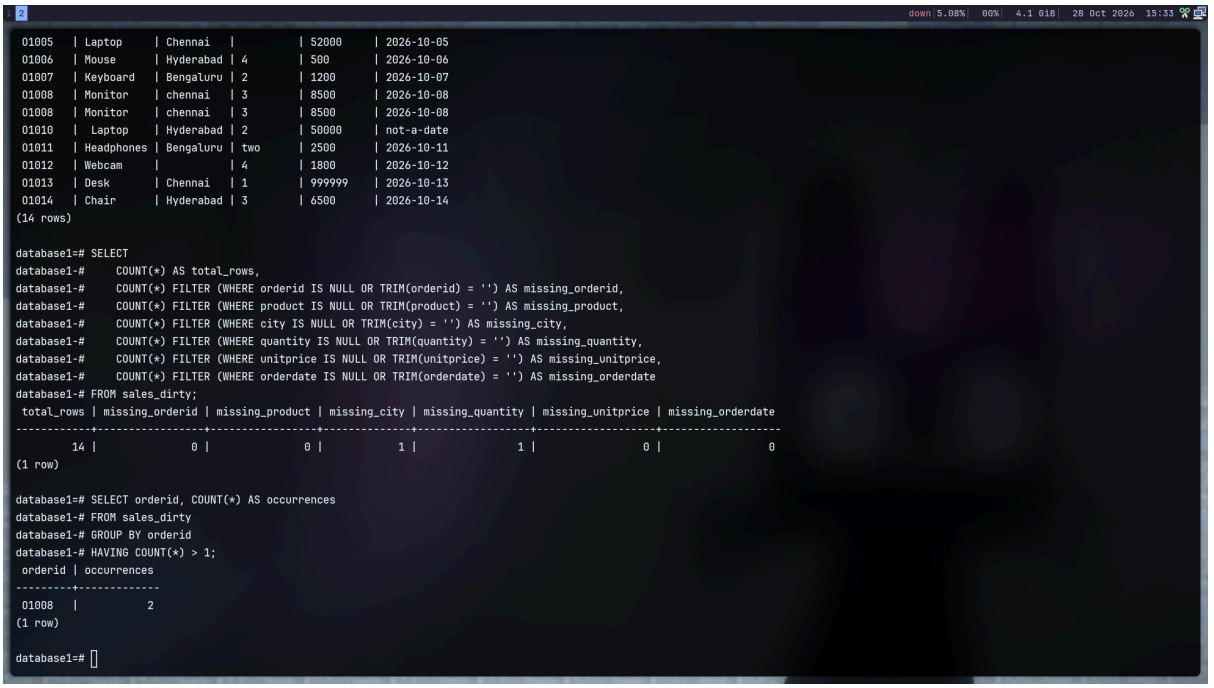
```
                                        ^
HINT:  Perhaps you meant to reference the column "sales_dirty.orderid".
database1=# select * from sales_dirty ORDER BY orderid;
 orderid |  product   |   city    | quantity | unitprice | orderdate
---------+------------+-----------+----------+-----------+------------
 01001   |  Laptop    | Chennai   | 2        | 50000     | 2026-10-01
 01002   | Mouse      | chennai   | 5        | 500       | 2026-10-02
 01003   | Keyboard   | CHENNAI   | 3        | 1200      | 2026-10-03
 01004   | Monitor    | Bengaluru | 2        | 8500      | 2026-10-04
 01005   | Laptop     | Chennai   |          | 52000     | 2026-10-05
 01006   | Mouse      | Hyderabad | 4        | 500       | 2026-10-06
 01007   | Keyboard   | Bengaluru | 2        | 1200      | 2026-10-07
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01010   |  Laptop    | Hyderabad | 2        | 50000     | not-a-date
 01011   | Headphones | Bengaluru | two      | 2500      | 2026-10-11
 01012   | Webcam     |           | 4        | 1800      | 2026-10-12
 01013   | Desk       | Chennai   | 1        | 999999    | 2026-10-13
 01014   | Chair      | Hyderabad | 3        | 6500      | 2026-10-14
(14 rows)

database1=# SELECT
database1-#     COUNT(*) AS total_rows,
database1-#     COUNT(*) FILTER (WHERE orderid IS NULL OR TRIM(orderid) = '') AS missing_orderid,
database1-#     COUNT(*) FILTER (WHERE product IS NULL OR TRIM(product) = '') AS missing_product,
database1-#     COUNT(*) FILTER (WHERE city IS NULL OR TRIM(city) = '') AS missing_city,
database1-#     COUNT(*) FILTER (WHERE quantity IS NULL OR TRIM(quantity) = '') AS missing_quantity,
database1-#     COUNT(*) FILTER (WHERE unitprice IS NULL OR TRIM(unitprice) = '') AS missing_unitprice,
database1-#     COUNT(*) FILTER (WHERE orderdate IS NULL OR TRIM(orderdate) = '') AS missing_orderdate
database1-# FROM sales_dirty;
 total_rows | missing_orderid | missing_product | missing_city | missing_quantity | missing_unitprice | missing_orderdate
------------+-----------------+-----------------+--------------+------------------+-------------------+-------------------
         14 |               0 |               0 |            1 |                1 |                 0 |                 0
(1 row)

database1=#
```

3. **Find duplicate order id’s**

   Query : SELECT orderid, COUNT(*) AS occurrences
   FROM sales_dirty
   GROUP BY orderid
   HAVING COUNT(*) > 1;



```
 01005   | Laptop     | Chennai   |          | 52000     | 2026-10-05
 01006   | Mouse      | Hyderabad | 4        | 500       | 2026-10-06
 01007   | Keyboard   | Bengaluru | 2        | 1200      | 2026-10-07
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01010   |  Laptop    | Hyderabad | 2        | 50000     | not-a-date
 01011   | Headphones | Bengaluru | two      | 2500      | 2026-10-11
 01012   | Webcam     |           | 4        | 1800      | 2026-10-12
 01013   | Desk       | Chennai   | 1        | 999999    | 2026-10-13
 01014   | Chair      | Hyderabad | 3        | 6500      | 2026-10-14
(14 rows)

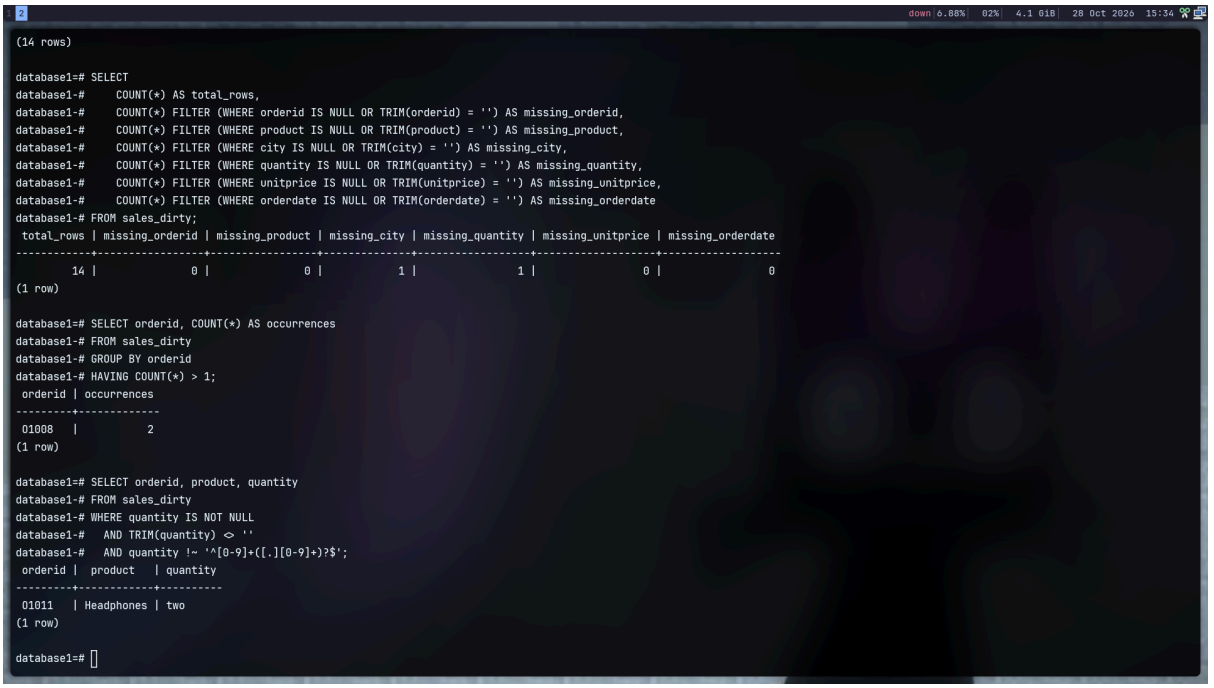
database1=# SELECT
database1-#     COUNT(*) AS total_rows,
database1-#     COUNT(*) FILTER (WHERE orderid IS NULL OR TRIM(orderid) = '') AS missing_orderid,
database1-#     COUNT(*) FILTER (WHERE product IS NULL OR TRIM(product) = '') AS missing_product,
database1-#     COUNT(*) FILTER (WHERE city IS NULL OR TRIM(city) = '') AS missing_city,
database1-#     COUNT(*) FILTER (WHERE quantity IS NULL OR TRIM(quantity) = '') AS missing_quantity,
database1-#     COUNT(*) FILTER (WHERE unitprice IS NULL OR TRIM(unitprice) = '') AS missing_unitprice,
database1-#     COUNT(*) FILTER (WHERE orderdate IS NULL OR TRIM(orderdate) = '') AS missing_orderdate
database1-# FROM sales_dirty;
 total_rows | missing_orderid | missing_product | missing_city | missing_quantity | missing_unitprice | missing_orderdate
------------+-----------------+-----------------+--------------+------------------+-------------------+-------------------
         14 |               0 |               0 |            1 |                1 |                 0 |                 0
(1 row)

database1=# SELECT orderid, COUNT(*) AS occurrences
database1-# FROM sales_dirty
database1-# GROUP BY orderid
database1-# HAVING COUNT(*) > 1;
 orderid | occurrences
---------+-------------
 01008   |           2
(1 row)

database1=#
```

4. **Find invalid quantities**

Query:SELECT orderid, product, quantity
FROM sales_dirty
WHERE quantity IS NOT NULL
  AND TRIM(quantity) <> ''
  AND quantity !~ '^[0-9]+([.][0-9]+)?$';



```
(14 rows)

database1=# SELECT
database1-#     COUNT(*) AS total_rows,
database1-#     COUNT(*) FILTER (WHERE orderid IS NULL OR TRIM(orderid) = '') AS missing_orderid,
database1-#     COUNT(*) FILTER (WHERE product IS NULL OR TRIM(product) = '') AS missing_product,
database1-#     COUNT(*) FILTER (WHERE city IS NULL OR TRIM(city) = '') AS missing_city,
database1-#     COUNT(*) FILTER (WHERE quantity IS NULL OR TRIM(quantity) = '') AS missing_quantity,
database1-#     COUNT(*) FILTER (WHERE unitprice IS NULL OR TRIM(unitprice) = '') AS missing_unitprice,
database1-#     COUNT(*) FILTER (WHERE orderdate IS NULL OR TRIM(orderdate) = '') AS missing_orderdate
database1-# FROM sales_dirty;
 total_rows | missing_orderid | missing_product | missing_city | missing_quantity | missing_unitprice | missing_orderdate
------------+-----------------+-----------------+--------------+------------------+-------------------+-------------------
         14 |               0 |               0 |            1 |                1 |                 0 |                 0
(1 row)

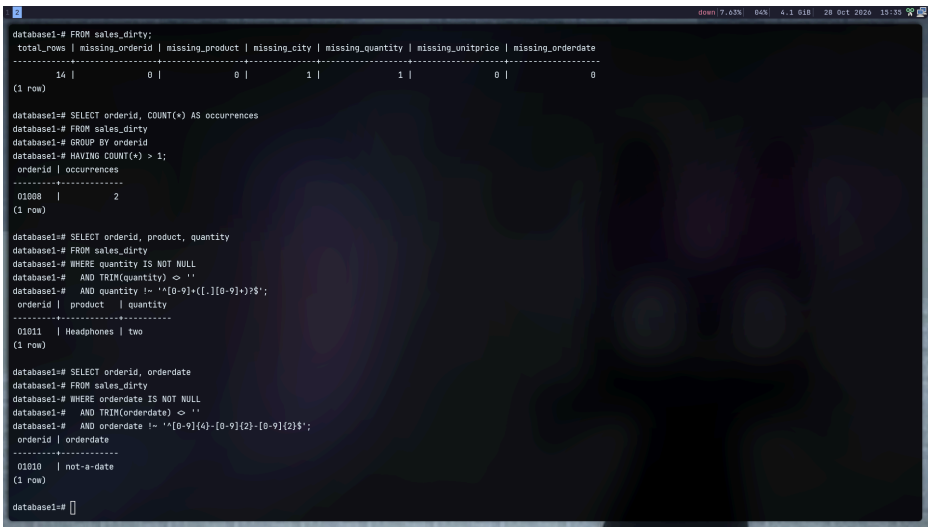
database1=# SELECT orderid, COUNT(*) AS occurrences
database1-# FROM sales_dirty
database1-# GROUP BY orderid
database1-# HAVING COUNT(*) > 1;
 orderid | occurrences
---------+-------------
 O1008   |           2
(1 row)

database1=# SELECT orderid, product, quantity
database1-# FROM sales_dirty
database1-# WHERE quantity IS NOT NULL
database1-#   AND TRIM(quantity) <> ''
database1-#   AND quantity !~ '^[0-9]+([.][0-9]+)?$';
 orderid |  product   | quantity
---------+------------+----------
 O1011   | Headphones | two
(1 row)

database1=#
```

5. **Invalid dates**

Query: SELECT orderid, orderdate
FROM sales_dirty
WHERE orderdate IS NOT NULL
AND TRIM(orderdate) <> ''
AND orderdate !~ '^[0-9]{4}-[0-9]{2}-[0-9]{2}$';



```
database1-# FROM sales_dirty;
 total_rows | missing_orderid | missing_product | missing_city | missing_quantity | missing_unitprice | missing_orderdate
------------+-----------------+-----------------+--------------+------------------+-------------------+-------------------
         14 |               0 |               0 |            1 |                1 |                 0 |                 0
(1 row)

database1=# SELECT orderid, COUNT(*) AS occurrences
database1-# FROM sales_dirty
database1-# GROUP BY orderid
database1-# HAVING COUNT(*) > 1;
 orderid | occurrences
---------+-------------
 O1008   |           2
(1 row)

database1=# SELECT orderid, product, quantity
database1-# FROM sales_dirty
database1-# WHERE quantity IS NOT NULL
database1-#   AND TRIM(quantity) <> ''
database1-#   AND quantity !~ '^[0-9]+([.][0-9]+)?$';
 orderid |  product   | quantity
---------+------------+----------
 O1011   | Headphones | two
(1 row)

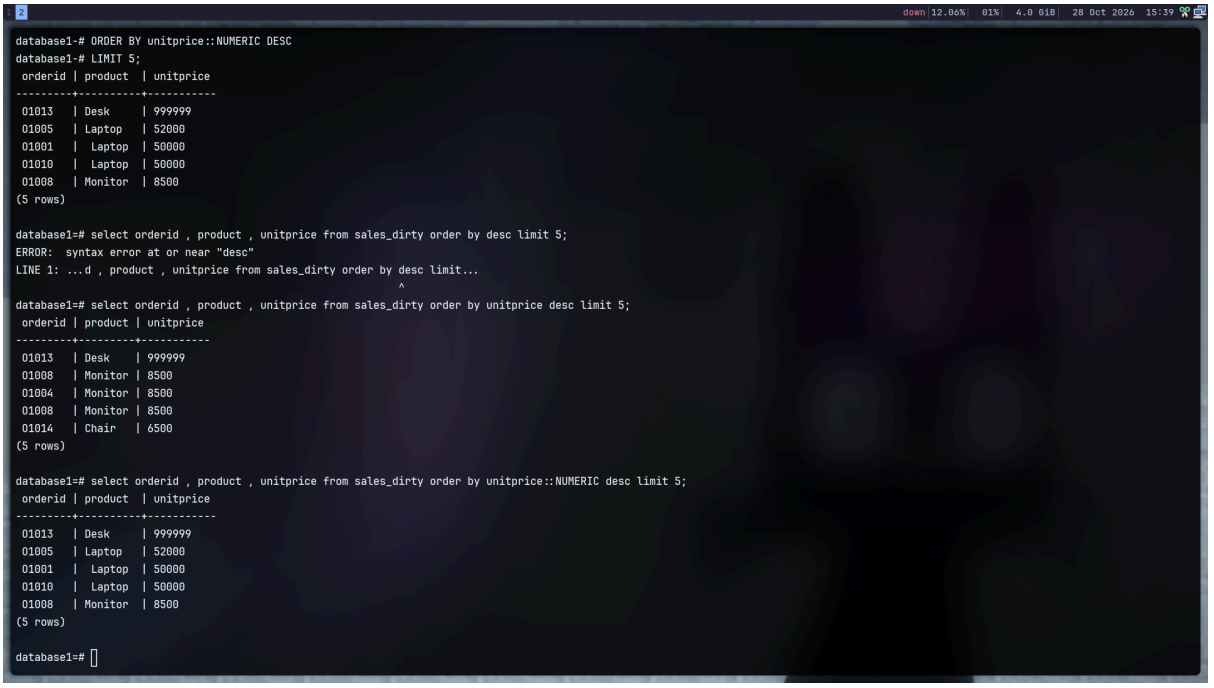
database1=# SELECT orderid, orderdate
database1-# FROM sales_dirty
database1-# WHERE orderdate IS NOT NULL
database1-#   AND TRIM(orderdate) <> ''
database1-#   AND orderdate !~ '^[0-9]{4}-[0-9]{2}-[0-9]{2}$';
 orderid | orderdate
---------+-----------
 O1010   | not-a-date
(1 row)

database1=#
```

6. **5 Highest unit price**

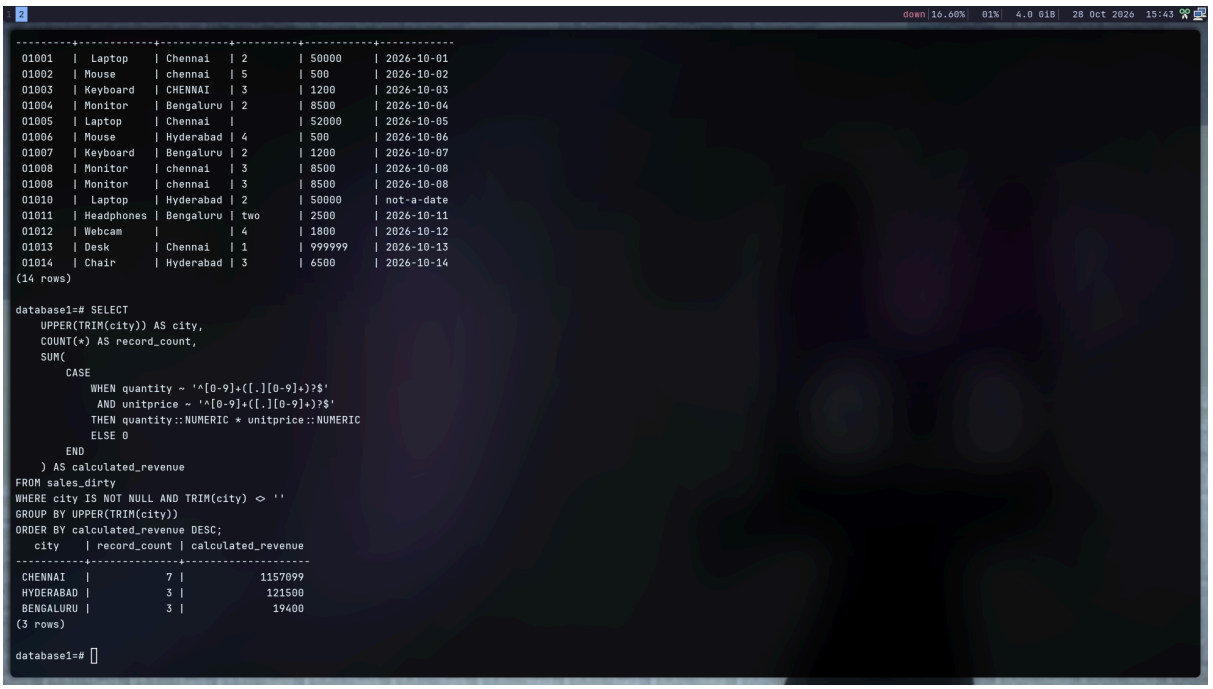
Query:

select orderid , product , unitprice from sales_dirty order by unitprice::NUMERIC desc limit 5;



7. **Compare Sales by each city**

```
SELECT
  UPPER(TRIM(city)) AS city,
  COUNT(*) AS record_count,
  SUM(
    CASE
      WHEN quantity ~ '^[0-9]+([.][0-9]+)?$'
       AND unitprice ~ '^[0-9]+([.][0-9]+)?$'
      THEN quantity::NUMERIC * unitprice::NUMERIC
      ELSE 0
    END
  ) AS calculated_revenue
FROM sales_dirty
WHERE city IS NOT NULL AND TRIM(city) <> ''
GROUP BY UPPER(TRIM(city))
ORDER BY calculated_revenue DESC;
```



```
 01001   |  Laptop    | Chennai   | 2        | 50000     | 2026-10-01
 01002   | Mouse      | chennai   | 5        | 500       | 2026-10-02
 01003   | Keyboard   | CHENNAI   | 3        | 1200      | 2026-10-03
 01004   | Monitor    | Bengaluru | 2        | 8500      | 2026-10-04
 01005   | Laptop     | Chennai   |          | 52000     | 2026-10-05
 01006   | Mouse      | Hyderabad | 4        | 500       | 2026-10-06
 01007   | Keyboard   | Bengaluru | 2        | 1200      | 2026-10-07
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01008   | Monitor    | chennai   | 3        | 8500      | 2026-10-08
 01010   |  Laptop    | Hyderabad | 2        | 50000     | not-a-date
 01011   | Headphones | Bengaluru | two      | 2500      | 2026-10-11
 01012   | Webcam     |           | 4        | 1800      | 2026-10-12
 01013   | Desk       | Chennai   | 1        | 999999    | 2026-10-13
 01014   | Chair      | Hyderabad | 3        | 6500      | 2026-10-14
(14 rows)

database1=# SELECT
    UPPER(TRIM(city)) AS city,
    COUNT(*) AS record_count,
    SUM(
        CASE
            WHEN quantity ~ '^[0-9]+([.][0-9]+)?$'
             AND unitprice ~ '^[0-9]+([.][0-9]+)?$'
            THEN quantity::NUMERIC * unitprice::NUMERIC
            ELSE 0
        END
    ) AS calculated_revenue
FROM sales_dirty
WHERE city IS NOT NULL AND TRIM(city) <> ''
GROUP BY UPPER(TRIM(city))
ORDER BY calculated_revenue DESC;
   city    | record_count | calculated_revenue
-----------+--------------+--------------------
 CHENNAI   |            7 |            1157099
 HYDERABAD |            3 |             121500
 BENGALURU |            3 |              19400
(3 rows)

database1=#
```

The record count represent how many times chennai have been in that column with standardizing the types like Chennai, CHENNAI , chennai etc…